BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. Which of the following is the basic structural and functional unit of ecology .....

(A) species
(B) ecosystem
(C) population
(D) community

Q2. External fertilization occurs in .....

(A) reptile
(B) birds
(C) amphibian
(D) mammals

Q3. The largest reservoir of nitrogen in our planet is .....

(A) ocean
(B) atmosphere
(C) phage DNA
(D) fossil fuels

Q4. Radiation therapy is a treatment of .....

(A) asthma
(B) lungs cancer
(C) bronchitis
(D) emphysema

Q5. Sedative drugs control the function of .....

(A) kidney
(B) lungs
(C) brain
(D) stomach

Q6. Most flowers have coloured .....

(A) sepals
(B) gynoecium
(C) petals
(D) stamens

Q7. Food is prevented from entering into larynx by .....

(A) glottis
(B) epiglottis
(C) vocal cords
(D) nostrils

Q8. Human have ..... pairs of spinal nerves.

(A) 12
(B) 24
(C) 31
(D) 38

Q9. Genes are short segments of .....

(A) proteins
(B) DNA
(C) carbohydrates
(D) lipids

Q10. The tube between kidney and urinary bladder is .....

(A) ureter
(B) urethra
(C) renal tubule
(D) nephron

Q11. AIDS is a ..... infection.

(A) fungal
(B) bacterial
(C) viral
(D) herbal

Q12. Enzymes and proteins may denature at temperature above .....

(A) 35°C
(B) 30°C
(C) 40°C
(D) 45°C

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Discuss oogenesis with diagram.

(ii) What do you mean by internal fertilization?

(iii) Why is pituitary gland called master gland?

(iv) Give a brief description of arthritis?

(v) Why did Mendel select peas as a subject of studying genetics?

(vi) Define ligaments and tendons?

(vii) State the functions of cerebrum.

(viii) Differentiate between population and community.

(ix) Define narcotics and give two examples.

(x) Describe continuous variation.

(xi) Write down any four effects of acid rain.

SECTION-C

Maximum Marks: 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Explain gametogenesis.

(ii) Describe the types of neurons on the basis of functions.

Q3.

(i) Explain the Mendel's law of segregation of genes.

(ii) What is crossing over and mutation?

Q4.

(i) Describe the major groups of antibiotics.

(ii) Write down the objectives of genetic engineering.

Q5.

(i) Differentiate between parasitism and mutualism.

(ii) Highlight the mechanism of breathing in human.